

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 29

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • The void fills with water naturally; wetlands are uncommon / valuable habitats in most regions (1) • The new habitat may have greater wildlife value than the farmland lost; low cost compared with filling and levelling (1)	2
01.2	<i>1 mark for each point.</i> • Housing: ground must be stable (no subsidence), no toxic metal wastes (otherwise industrial use is safer) (1) • Farmland: landscape not too steep or uneven, no toxic materials, sufficiently fertile soil (topsoil replaced) (1)	2
01.3	<i>1 mark.</i> • Species that cannot be caught / found, or whose habitat requirements are unknown or hard to recreate, are lost; transplanted plants often fail (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 140, 142; AQA spec 3.2.3.6.	
02.1	<i>1 mark for each point.</i> • Surplus energy drives a pump compressing air stored in underground caverns (salt mines); releasing it drives machinery / turbines (1) • Compression produces a lot of heat; storing this heat (hot oil / molten salt) and returning it when the air expands recovers energy that would otherwise be lost (1)	2
02.2	<i>1 mark for each point.</i> • As volume increases the surface area : volume ratio decreases (1) • Heat is lost through the surface so there is less surface per unit of stored heat; very large stores can carry summer solar heat into winter (inter-seasonal storage) (1)	2

Question	Marking guidance	Marks
02.3	<i>1 mark for each point.</i> - Potassium nitrate has a much higher boiling point so it stores heat at up to 550 °C without pressurisation (1) - Water could only reach such temperatures under high pressure, increasing costs; high temperature needed to raise steam for turbines (1)	2
02.4	<i>1 mark for each point.</i> - Materials with high specific heat capacity (concrete, water) warm and cool slowly (1) - They absorb heat in hot weather reducing overheating / air conditioning, and release it as the weather cools, delaying the need for heating (1)	2
02.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Flywheels: rotating mass stores kinetic energy for short-term smoothing / generation (1) - Supercapacitors: electrochemical capacitors with 10-100 times the energy density of conventional capacitors; possible future large-scale electricity storage (1) - Thermal storage in molten salt / high-volume tanks; hydrogen / P2G; batteries (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 222-224; AQA spec 3.3.4.1.	
03.1	<i>1 mark for each point.</i> - Fission: splitting the nuclei of large atoms (U-235, Pu-239) when hit by neutrons (1) - Fusion: joining the nuclei of small atoms (deuterium and tritium - isotopes of hydrogen) to form helium, releasing a neutron and energy - the energy source of stars (1)	2
03.2	<i>1 mark for each point.</i> - Deuterium (hydrogen-2) extracted from (sea)water (1) - Tritium (hydrogen-3) produced by neutron bombardment of lithium (a lithium blanket around the reactor) (1)	2
03.3	<i>1 mark for each point.</i> - Very high temperature turns hydrogen into a plasma (electrons stripped away) and gives nuclei enough kinetic energy to overcome the repulsion between positive nuclei and collide (1) - Vacuum so the plasma is not cooled by air (1) - Magnetic field holds the plasma centrally so it does not touch the container walls and cool (1)	3
03.4	<i>1 mark for each point.</i> - Fuel (deuterium / lithium) is abundant / effectively unlimited (1) - Much less radioactive waste (no long-lived fission products); no chain reaction / meltdown risk; no weapons material (1) - Not yet achieved sustained net energy output: reactors (JET) use more energy than they release; ITER aims for 500 MW out from 50 MW in; laser fusion also experimental; commercial viability decades away (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 195-196; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> • Spearman's rank correlation coefficient (1) • Mann-Whitney U test (1) • Chi-squared test (1) • t-test (1)	4
04.2	<i>1 mark for each point.</i> • Chi-squared compares frequencies / counts in categories (1) • Dissolved oxygen is continuously variable measured data, so a t-test or Mann-Whitney U test is needed (1)	2
04.3	<i>1 mark for each point.</i> • There is always some remaining uncertainty / a probability that the result was produced by chance (1) • Statistics only give a level of confidence (eg 95%, 99%) that the result is significant (1)	2
04.4	<i>1 mark for each point.</i> • Bars drawn above and below each mean showing plus and minus one standard deviation (1) • If the bars of two means overlap a lot the difference is probably not significant; if they do not overlap it probably is (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	
05.1	<i>Correct answer only.</i> • Aerobic decomposition produces little methane and returns nutrients to the soil (1)	1
05.2	<i>1 mark for each point.</i> • It may contain concentrated heavy metals / toxic residues (1) • It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble (1)	2
05.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Impermeable polymer / clay liner to stop leachate escaping (1) • Collection and treatment of leachate (1) • Collection of methane and use as fuel (1) • Daily covering of waste / final capping and restoration (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	
06.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Water = $65.0 - 55.0 = 10.0$ g; fresh soil = 40.0 g (1) • $10.0 / 40.0 \times 100 = 25.0\%$ (1)	2
06.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Organic matter = $55.0 - 52.6 = 2.4$ g; dry soil = 30.0 g (1) • $2.4 / 30.0 \times 100 = 8.0\%$ (1)	2
06.3	<i>1 mark for each point.</i> • So all water is removed and the mass loss on burning is due to organic matter only (1) • Heating until mass no longer changes shows all water has evaporated (1)	2

Question	Marking guidance	Marks
06.4	<i>1 mark for each point.</i> - Organic matter releases nutrients as it decomposes and provides food for soil biota (1) - Increases water retention and, as humus, binds particles into crumb peds improving structure / aeration (1) - Surface organic matter / humus adhesion reduce rain splash and wind erosion (1)	3
06.5	<i>1 mark.</i> Samples taken from different depths / locations (organic matter is concentrated near the surface); incomplete burning; different heating times (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-159, 412; AQA spec 3.2.5.1, 3.7.2.1.	
07.1	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> - Sludge from primary sedimentation is digested by anaerobic microbes in a warm tank for about four weeks (1) - Kills most pathogens (1) - Reduces odour and volume of the sludge (1) - Produces biogas (mainly methane with some CO₂) used as fuel to heat the digester / generate electricity (1) - Digested sludge has a better C:N ratio and can be used as agricultural fertiliser (1)	4
07.2	<i>1 mark for each point.</i> - Heavy metals from urban wastes / road runoff may be present and absorbed by crops - reduced by adding lime to lower metal solubility (1) - Remaining pathogens - destroyed by heat treatment of the sludge before use (1) - Timing: spreading in the growing season covers crops; in winter vehicles compact wet soils - apply at suitable times / when soils are dry (1) - Odour / public objection - digestion reduces odour (1)	4
07.3	<i>1 mark for each point.</i> - Foul waste is a small volume needing full treatment so smaller, cheaper treatment works can be built; surface water needs little treatment (1) - The treatment works is not affected by weather-related fluctuations in flow (storms) so storm overflows of untreated sewage are avoided (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 270-273; AQA spec 3.4.3.2.9.	
08.1	<i>1 mark for each point.</i> - Most solar energy is absorbed at the surface in the tropics where insolation is most intense / the Sun is overhead (1) - The warm surface heats the air above; warm winds (eg south-westerlies) carry heat energy to higher latitudes (1) - Winds blowing over the oceans create currents (North Atlantic Conveyor / Gulf Stream) carrying warm water from the tropics to NW Europe (1) - Evaporation in the tropics stores latent heat in water vapour which is released when it condenses at higher latitudes; also transports water to areas that would be dry (1)	4

Question	Marking guidance	Marks
08.2	<i>1 mark for each point.</i> - The North Atlantic Conveyor brings warm water from the tropics to the coast of NW Europe (1) - Prevailing south-westerly winds blowing over the warm sea carry the heat onto the land (1)	2
08.3	<i>1 mark for each point.</i> - Atmospheric pressure controls how easily water molecules evaporate / escape from the surface (1) - At low pressure water would boil / evaporate at low temperatures so all surface water would become vapour (1)	2
08.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Nitrogen (1) - Oxygen (1) - Carbon dioxide (1) - Argon / neon / krypton / xenon (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87, 96-97; AQA spec 3.2.1.1, 3.2.2.3.	
09.1	<i>1 mark for each point.</i> - Manufacture and use of CFCs and other ozone-depleting substances (ODSs) phased out then banned (1) - HCFCs to be phased out (by 2030); some essential uses (eg halon fire extinguishers in aircraft) still permitted (1) - A fund to help (developing) countries implement the Protocol; legally binding targets (unlike the 1985 Vienna Convention) (1)	3
09.2	<i>2 marks for each point (partial credit for a partly correct point).</i> - Aerosols: pump-action sprays / roll-on deodorants (alternative process), or hydrocarbon propellants (propane, butane) - flammable / fire risk (2) - Refrigerants: HCFCs (less stable so break down in the troposphere, but still some ozone depletion) then HFCs (no chlorine) - more expensive, less effective, powerful greenhouse gases (2)	4
09.3	<i>1 mark for each point.</i> - CFCs are drained from the appliance and incinerated / broken down at high temperature (1) - Products are CO₂ and acidic gases (HCl, HF) which are neutralised with crushed lime (1)	2
09.4	<i>1 mark.</i> CFCs are persistent, so those already released continue to reach the stratosphere and destroy ozone for many years; some ODSs (eg dichloromethane) are not banned (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 112-113; AQA spec 3.2.1.3.	
10.1	<i>1 mark for the correct answer.</i> 16.0 - 5.0 = £11 million (1)	1
10.2	<i>1 mark for each point.</i> - It is rarely practical to reduce emissions to zero and the impact of low-level emissions may be acceptable, so an acceptable level must be decided (1) - ALARA: emissions should be As Low As Reasonably Achievable (1) - BATNEEC: the Best Available Technology Not Entailing Excessive Cost is chosen to achieve this (1)	3

Question	Marking guidance	Marks
10.3	1 mark for each point. - As industries expand or cities grow the number of pollutant sources increases (1) - So the total emissions / combined concentration rises even if each source is unchanged, and higher standards are needed (1)	2
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Critical Pathway Analysis: predicting movement of mobile persistent pollutants (state of effluent, wind, currents, geology, pH / oxygen and solubility, bioaccumulation, food sources, half-lives) so monitoring focuses on sites at risk; mainly radioactive discharges; checked by sampling - Critical Group Monitoring: exposure of the most-exposed group monitored before health effects; if their risk is acceptable everyone else is safer; emissions reduced if not - Emission control strategies: location (marine discharges, downwind of cities, not over aquifers) and timing (avoiding temperature inversions, low river flow) - Principles: polluter pays (incentive depends on penalties and enforcement), precautionary principle (assume harm until shown otherwise; ignorance no excuse), ALARA / BATNEEC and the non-linear cost-efficiency relationship - Selection of methods: production prevention (desulfurisation), prevention of release (precipitators, catalytic converters), post-release remediation (oil spill clean-up, phytoremediation), alternative processes (electric vehicles, renewables, pyrethroids) - Evaluation: prevention usually cheaper and more effective than remediation; costs rise steeply near 100%; CPA relies on understanding of pathways; enforcement and international agreement needed for transboundary pollutants; judgement about acceptable risk is political as well as scientific - Justified conclusion	9
	Model answer - Level 3 (8 marks)	

Question	Marking guidance	Marks
	Deciding how much to control pollution starts with predicting where a pollutant will go. Critical Pathway Analysis uses the state of the effluent, wind and currents, geology, the effect of pH and oxygen on solubility, bioaccumulation and local food sources to predict whether a discharge will disperse harmlessly or concentrate somewhere sensitive; it is mainly used for radioactive discharges and means monitoring can focus on the sites at risk rather than sampling everywhere. Critical Group Monitoring then checks the exposure of the people most at risk from their lifestyle before any health effects occur: if their risk is acceptable everyone else's is lower, and if not, emissions are reduced.	
	The choice of control is guided by principles. The polluter pays principle gives a financial incentive to prevent damage, but only if penalties are high and polluters cannot escape responsibility. The precautionary principle assumes a waste is harmful until research shows otherwise, so ignorance is no defence. Because the cost of control rises steeply as emissions approach zero, emissions are set As Low As Reasonably Achievable using the Best Available Technology Not Entailing Excessive Cost. Where emission location and timing can be chosen (marine outfalls, downwind of towns, not over aquifers; avoiding temperature inversions) severity is reduced cheaply.	
	Overall, prevention of production (fuel desulfurisation) or of release (electrostatic precipitators, catalytic converters) is more effective than post-release remediation such as oil-spill clean-up, which never recovers all the pollutant. CPA depends on understanding pathways, and transboundary pollutants need international enforcement, but together these principles give a rational basis for deciding acceptable risk rather than aiming for an unaffordable zero.	
	Weaker answer - Level 1 (3 marks)	
	Pollution can be controlled in lots of ways. Critical Pathway Analysis looks at where pollution goes and Critical Group Monitoring looks at people. The polluter pays principle means the person who pollutes has to pay. The precautionary principle means being careful. Pollution can be prevented, or cleaned up afterwards with things like booms for oil spills. Electric cars are an alternative process. Pollution control can be expensive so ALARA is used. All of these are good ways of controlling pollution.	
	What makes the difference	
	• The Level 3 answer explains <i>how</i> each principle works (why penalties matter, why costs rise near zero emissions); the weak answer only names them.	
	• The Level 3 answer groups the material logically (predict, decide, select) rather than listing everything at the same weight.	
	• 'Evaluate' needs comparison and a judgement: prevention vs remediation, CPA's dependence on understanding pathways, and a conclusion about acceptable risk. The weak answer ends with 'all of these are good'.	
	• Precise examples (desulfurisation, precipitators, catalytic converters, temperature inversions) show applied knowledge; 'things like booms' does not.	

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
11.1	Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: toxicity (LD50, MDAF, MDNF), specificity, persistence, liposolubility/water solubility, mobility, mode of action (contact/systemic), synergism • Bioaccumulation and biomagnification explained by persistence + liposolubility; DDT and birds of prey as a case study • Comparison of organochlorines, organophosphates, pyrethroids, neonicotinoids with their relative advantages and disadvantages • Reducing impact: restrictions/bans, choice of non-persistent and specific pesticides, systemic pesticides and seed coatings, application timing (night, still days, not in flower), droplet size • Alternatives: cultural control (rotation, barrier/companion crops), biological control, predator habitats, sterile males, pheromone traps, genetic resistance, GM crops, integrated pest management • Trade-offs: yield, cost, human health (organophosphates), resistance evolution, pollinator services; risk:benefit; sustainability and the circular economy • Judgement: no pesticide is perfectly specific; combination of property-based selection and integrated control is most sustainable Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240, 262-266, 314-320; AQA spec 3.4.3.2.8, 3.5.1.1, 3.5.1.5.	25

Question	Marking guidance	Marks
11.2	To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society? Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Fossil fuel features that shaped society: energy density, storage, transport, high temperatures, liquid vehicle fuels, established infrastructure • Renewable features: intermittent / unpredictable (wind, solar) vs predictable (tidal) vs controllable (HEP, biofuel, geothermal); low energy density; location; low pollution but manufacturing impacts • Technologies and developments for each resource with examples; new technologies improving usability • Storage and grid management: pumped storage, batteries, hydrogen economy, P2G, V2G, thermal storage, peak shaving; the role of electricity as a secondary fuel • Economics and politics: development costs, interest, subsidies, external costs of fossil fuels, falling renewable costs • Energy conservation as part of the solution; sustainability principles - natural systems run on low-density renewable energy • Judgement: electricity largely yes; heat, aviation, shipping and high-temperature industry harder; transition period with nuclear / CCS Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25